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PABLO BARBECHO BAUTISTA, Polytechnic University of Catalonia, Barcelona, Barcelona, Spain

LUIS F URQUIZA-AGUIAR, National polytechnic school, Quito, Pichincha, Ecuador

MÓNICA AGUILAR IGARTUA, Polytechnic University of Catalonia, Barcelona, Barcelona, Spain

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Published: 16 November 2020

[Citation in BibTeX format](#)

MSWiM '20: 23rd International ACM Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems

November 16 - 20, 2020
Alicante, Spain

Conference Sponsors:
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Evaluation of Dynamic Route Planning Impact on Vehicular Communications with SUMO

Pablo Barbecho Bautista*
Department of Network Engineering
Universitat Politècnica de Catalunya
(UPC)
Barcelona, Spain
pablo.barbecho@upc.edu

Luis Urquiza-Aguilar
Dept. Electrónica,
Telecomunicaciones y Redes de
Información.
Escuela Politécnica Nacional
Quito, Ecuador
luis.urquiza@epn.edu.ec

Mónica Aguilar Igartua
Department of Network Engineering
Universitat Politècnica de Catalunya
(UPC)
Barcelona, Spain
monica.aguilar@upc.edu

ABSTRACT

Simulations are the first approach used by the research community to evaluate mobile ad hoc networks. Particularly, vehicular ad hoc networks (VANETs) are a singular type of mobile ad hoc networks that raise technical challenges, for instance, in the context of vehicular mobility models. When assessing VANETs, realistic vehicular models are essential to produce meaningful evaluation results. In this context, realistic vehicles' mobility includes re-routing capabilities that allow vehicles to re-compute their routes in front of specific traffic conditions (e.g., traffic jams). In this paper, we provide a thorough analysis of the influence of enabling re-routing properties on (i) the mobility of the vehicle and (ii) on the connections of the vehicular network. For this, we use the road traffic simulator SUMO to generate vehicular traces, and then we will analyze the connectivity of the vehicular network employing well-known graph metrics. Our results provide insights about the behavior of the vehicle's mobility and the nodes' connectivity.

CCS CONCEPTS

• **Networks** → **Network performance evaluation**; Mobile ad hoc networks.

KEYWORDS

Automatic routing, vehicular traffic, SUMO, VANET

ACM Reference Format:

Pablo Barbecho Bautista, Luis Urquiza-Aguilar, and Mónica Aguilar Igartua. 2020. Evaluation of Dynamic Route Planning Impact on Vehicular Communications with SUMO. In *23rd International ACM Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems (MSWiM '20)*, November 16–20, 2020, Alicante, Spain. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3416010.3423233>

*Pablo Barbecho Bautista and Luis Urquiza-Aguilar contributed equally to this research.

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MSWiM '20, November 16–20, 2020, Alicante, Spain

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ACM ISBN 978-1-4503-8117-8/20/11...\$15.00
<https://doi.org/10.1145/3416010.3423233>

1 INTRODUCTION

Nowadays, simulation is the first approach researchers follow to assess the performance of vehicular networks. While the deployment of real test scenarios is infeasible in most cases (they are complex and costly), simulation techniques are commonly used to study vehicular network protocols and services for intelligent transportation systems (ITS). In the context of vehicular networks, the simulation of inter-vehicle communications typically involves the coupling of a network simulator (e.g., Veins, NS-3) and a realistic road traffic simulator (e.g., VISSIM, DRACULA, SUMO), including events detection (e.g., traffic jams, accidents), mobility models, realistic maps, and route planning capabilities.

A wide-spread road traffic simulator is Simulation of Urban MObility (SUMO) [7]. SUMO is an open-source space-continuous road traffic simulator commonly used for testing VANETs and ITS services. SUMO allows users to handle large road traffic networks. It includes components of the road network and vehicular demand modeling (e.g., traffic lights, right-of-way rules, lane changing) and public transport and pedestrians. Here, vehicles' traces can be defined manually or using tools included in the SUMO package (e.g., DUArouter, MArouter, OD2trips, Randomtrips) [13].

Most modern vehicles are equipped with real-time navigation services, allowing drivers to better use the available road capacity. Besides, drivers can be informed through the vehicular network about the road network's events during their trips. In this context, most drivers are familiarized with the road network (urban environment) and, therefore, can decide to try an alternative path during their trips. This behavior is emulated in SUMO utilizing an additional feature, called *re-routing device*, with which vehicles may be equipped. If vehicles are equipped with a re-routing device, they can change their current routes, taking into account the road network's current state. In this sense, SUMO includes capabilities for re-routing vehicles dynamically. This approach impacts the overall road traffic conditions (i.e., the road congestion level). Consequently, the data network connectivity in the VANET is also affected by these vehicles' re-routing scheme.

Many researchers put their effort into the vehicular network model and the validation of results without considering the traffic state's effect on the road network during the simulation. In this paper, we investigate the impact of enabling vehicles' re-routing capabilities (i) over the vehicles' mobility, and the effect (ii) on the connectivity of the vehicular network. Using the popular traffic simulator SUMO, we simulate the traffic mobility and the automatic vehicles' re-routing property on a real map. Then, to evaluate

the inter-vehicle connectivity, we take advantage of graph theory concepts. First, we perform an offline analysis transforming the vehicular traces into snapshots at different simulation times to latter evaluate the network employing well-known graph metrics. The remainder of the paper is organized as follows: Section 2, presents some related work. In Section 3 the simulation environment is described, highlighting the automatic re-routing options. In Section 4, we investigate the effect of vehicles' re-routing on the vehicles' mobility and the nodes' connectivity. Finally, in Section 5 conclusions and future perspectives are presented.

2 RELATED WORK

Over the last few years, the vehicular networking community has proposed many proposals, applications, and services. One recurrent problem is to develop a reliable and usable mobility scenario to evaluate services for intelligent transportation systems (ITS). In this context, several works present realistic simulation scenarios derived from real traffic statistics and mobility patterns.

Regarding different scenarios available to the research community in [4], a realistic vehicular scenario is proposed called *Luxembourg SUMO Traffic* (LuST) scenario. Codecá. et al. built LuST from the road network of a real city imported from OpenStreetMaps (OSM) [10], including traffic demands and mobility patterns. In that work, the authors use traffic statistics of Luxembourg city to calibrate the traffic demand. Then, traffic demand was generated using the ACTIVITYGEN SUMO tool [11] based on the population and considering everyday trips (e.g., bus lines). The work includes an analysis of the impact of the traffic demand within 24 hours. Also, in [2], mixed traffic analysis of autonomous vehicles (AVs) and classic vehicles is presented. In that work, authors use linear approximations to calibrate and validate SUMO models based on real-traffic measurement data. Then, the effects of varying penetration rates of AVs are analyzed in terms of traffic flow characteristics.

In contrast to previous works in [5], the Monaco SUMO Traffic (MoST) scenario includes a multimodal scenario that integrates vulnerable users intending to study advanced parking management solutions. It is based on synthetic activity-based mobility traces. The authors present mobility traces for pedestrians and motorbikes.

Concerning the vehicular network connectivity analysis, in [6], authors propose a realistic capacity analysis for VANETs through graph theory concepts. The proposed network model is constructed utilizing a Euclidean planar graph and an interference relationship graph. First, the authors modeled urban area structures as grid-based structures using a Euclidean graph, where vertices represent points in the plane, and the Euclidean distance between those points means the edges. Then, the interference relationship graph shows the interference relationship between every two units abstracted from the Euclidean planar graph.

In addition to the considerations typically used in the literature (real maps, real traffic demand), we consider the effect of enabling dynamic vehicle routing. On the one hand, we claim that by allowing vehicles' re-routing capabilities, the congestion levels of the scenario's vehicular traffic are significantly affected. On the other hand, the effect of vehicles' dynamic route planning modifies the vehicular network behavior.

3 SIMULATION ENVIRONMENT

In this section, we describe the methodology steps followed to generate both the *road network* and the *traffic demand*; see the flowchart depicted in Fig. 1. By *road network*, we refer to the traffic-related part of a map, including roads (edges), traffic lights, intersections, and the logic between elements. By *traffic demand*, we mean the number of vehicles circulating in the road network at a given time of day (e.g., peak hours, off-peak hours). The first step is to build

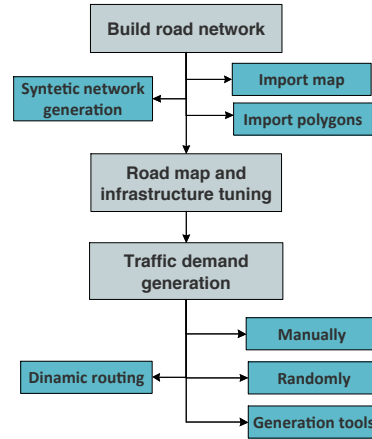


Figure 1: Flow chart followed to generate the road network and the traffic demand.

a road network. By using SUMO built-in tools, two methods can be followed: (i) create a synthetic network (e.g., a grid network), or (ii) generate a realistic network based on real maps. In this work, we follow the (ii) approach and import a map from OSM repositories [10]. Here, OSM excels other repositories because it is free and presents an excellent real-world representation.

To generate the road network, SUMO provides several tools [17]. In this work, we have used the following:

Netconvert: It takes digital map files from different sources (e.g., OSM, VISSIM, MATsim) and converts them into one road network file readable by SUMO, called *.net* file. It provides a full set of processing options for map conversion. A complete list of options is available in [16]. Common options to characterize urban environment maps, are *remove-edges.by-vclass*, *remove-edges.isolated*, *no-turnarounds*, and *geometry.remove*, as it was detailed in a previous work [20].

Polyconvert: It allows us to import geometrical shapes (polygons) from different sources (e.g., OSM). It is particularly important when evaluating vehicular networks operating in suburban and urban environments. Due to the presence of buildings (i.e., polygons), radio transmissions are heavily impacted by signal shadowing effects in this environment.

Once the road network has been defined, to further modify or debug network properties (e.g., roads' length, traffic lights, intersections logic), the SUMO package includes a useful graphical tool:

Netedit: It is a visual network editor that allows us to edit/create all available network properties and infrastructure in the network (e.g., bus stops, traffic light cycles).

Then, the traffic demand generation, see Fig. 1, refers to the definition of the vehicles circulating on the road network during the simulation. In SUMO, each vehicle is defined by a vehicle type that describes the vehicle's physical properties (e.g., size, gas/electric fueled), a route the vehicle shall take, and the vehicle itself. A single route can be assigned to a single-vehicle (e.g., trips, routes) or several vehicles (e.g., flows of vehicles). At this point, routes can be defined in three different ways, see Fig. 1: (i) manually specify all the components of the route (i.e., set of edges), (ii) generate routes randomly, or (iii) use the SUMO generation tools. In this work, we use (ii) to randomly create routes for the road network. For this, the following tools are used:

DUARouter: It computes the shortest path between a vehicle's origin and destination in terms of trip time or trip length. Different demand definitions, such as trips or flows, can be used as input of the script. Several routing algorithms can be configured to find a new vehicle's route (e.g., Astar, Dijkstra, etc.). By default, *duarouter* uses the Dijkstra algorithm [15].

RandomTrips.py: This script allows us to quickly generate a set of random trips on a given road network, which is required as an input parameter for the script [19]. By default, the source and destination of the route are selected randomly. As the destination point is randomly selected, it may not be reachable from the origin point. The option *-r* automatically calls *duarouter* [15] tool to discard disconnected trips.

Finally, vehicles' routes are composed of a set of edges. Then, vehicles are denoted by the combination of a vehicle type and a given route. By default, in SUMO, each vehicle will circulate along the edges of its corresponding path. Following this approach, some realistic route events (e.g., traffic jams) during the simulation will not be tackled. That is, if a road becomes congested, vehicles do not modify their routes as it would be desirable in a realistic scenario. Dynamic route planning can be enabled in SUMO to cope with these situations. Here, vehicles can modify their routes during the simulation [12], which reflects what would happen in reality. In the following section, re-routing configurations are detailed.

3.1 Dynamic route planning

Nowadays, most modern vehicles are equipped with real-time navigation services, allowing drivers to plan their routes before and during their trip. Vehicles need to change their routes during their trips dynamically to provide a realistic mobility pattern. In SUMO, each vehicle can be equipped with a re-routing device allowing vehicles to change their routes during their trips dynamically. A device refers to a container for data and functionality that enable vehicles to interact with their environment.

This re-routing approach works by giving vehicles the ability to re-compute their routes periodically (i.e., re-routing is triggered by the vehicle). For this, some or all the vehicles in the simulation can be equipped with a re-routing device [14]. By default, the percentage of vehicles that have a re-routing device is set to 0. Four main steps are considered to configure vehicles' re-routing:

Step 1: Before the simulation initializes, each vehicle has assigned a vehicle type and a route, as it is detailed in Section 3. Also, the percentage of vehicles equipped with a re-routing device is defined.

Step 2: Once the simulation starts, conditions along the road network are updated following a fixed time interval of t (seconds).

For this, SUMO assigns a weight to each edge in the network. The evolution of the edges' weights during simulation is computed as an estimation of the travel time (time required to traverse an edge). The travel time is estimated by calculating the mean speed on the road and then dividing the road length by that speed.

Step 3: Vehicles equipped with a re-routing device re-apply the optimal path computation following the r (seconds) re-routing interval. r refers to the interval time elapsed between consecutive re-routing events. If the travel time on the new route is shorter than on the current one, the vehicle changes the route accordingly.

Step 4: At this time, to select the fastest path for the current vehicle, SUMO uses the vehicle's current position, destination's position, and current status in the road network (updated at t time). If the new path results faster than the current one, the vehicle updates its route. The fastest path is computed using the *DUARouter* tool, which is commented in Section 3.

In the next section, we first take the mobility traces generated in Section 3, and then we vary the parameters associated with the vehicles' re-routing. The goal is to analyze the mobility pattern's impact and on the nodes' connectivity (vehicular network).

4 RESULTS ANALYSIS

We divide the analysis into two parts. First, we study the impact of automatic routing on traffic mobility employing trip statistics. Here, we consider fundamental mobility metrics (trip time and distance) and the number of route changes performed by vehicles during the simulation. We then study the impact of automatic routing in the vehicular communication network by using graph theory concepts. For this, we evaluate the communication between nodes utilizing well-known graph metrics [8].

4.1 Simulation settings

In this work, we use a realistic scenario, shown in Fig. 2, to carry out the simulations. It corresponds to Gracia's district and its surroundings in the city of Barcelona, Spain, with an area of 7 km². The road network has been generated using a real map imported from OSM [10] through the use of the *Netconvert* SUMO tool, described in Section 3. Also, roads' speed information has been imported from OSM; see the colors bar in Fig. 2. Exclusive lanes for the tram, rail, electric rail, bicycles, and pedestrians have been removed from the road network, as part of the road map tuning process, according to Fig. 1. Besides, buildings information has been generated by using the *Polyconvert* tool described in Section 3. Also, the area of study has been tuned using the *Netedit* tool (i.e., roads were set within the area of study).

Once the road network has been defined, we generated mobility traces. We consider data related to real traffic demand in the district of Gracia, according to Barcelona's City Hall [1]. The traffic demand refers to the mean number of vehicles per hour that circulate on a working day. For this area, in rush hours, there are around 17500 *vehicles/hour*. In Table 1, we have derived the number of vehicles for 40%, 70%, and 100% of the maximum real traffic demand; henceforth low, medium, and high vehicular traffic demand, respectively. We have generated vehicles for three hours for each traffic demand through the use of the *RandomTrips.py* tool in combination with the *DUARouter* tool, described in Section 3.

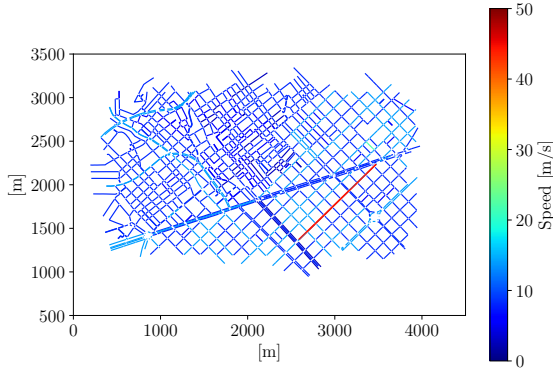


Figure 2: Simulation scenario exported from OSM [10]. Gracia district of Barcelona (Spain) with an area of 7 km². Maximum allowed speed per road is depicted in m/s.

Table 1: Number of vehicles generated during three hours (working day) according to real traffic in the district of Gracia (Barcelona, Spain), with an area of 7 km² [1].

Hours	40%	70%	100%
1	6981	12280	17591
2	7014	12272	17497
3	6973	12259	17457
Total	20968	36811	52545

We implemented the dynamic route planning using a re-routing device, as detailed in Section 3.1. A vehicle equipped with a re-routing device will update its route according to the re-routing interval of r (seconds). We consider the following scenarios:

- NR: None of the vehicles is capable of changing its route (default SUMO configuration).
- R - 1 min: All the vehicles in the simulation can change their routes following a re-routing interval of $r = 1$ min.
- R - 3 min: All the vehicles in the simulation can change their routes following a re-routing interval of $r = 3$ min.
- R - 6 min: All the vehicles in the simulation can change their routes following a re-routing interval of $r = 6$ min.

4.2 Traffic mobility analysis

SUMO provides several statistics [18] from simulations. We consider fundamental traffic measures: (i) the trip time, defined as the minimum time required for the vehicle to travel from origin to destination; (ii) the trip length, defined as the total length from origin to destination; and (iii) the number of route changes each vehicle performs. Note that the a vehicle will update its route if it finds a new path to the destination faster than the current one, as detailed in Section 3.1.

Trip time: Fig. 3 shows the improvement in the travel time by enabling re-routing capabilities. The most significant improvement can be observed for the "R - 1 min" case with the 40% of the traffic demand, see Fig. 3(a). By the end of the fourth hour, vehicles require

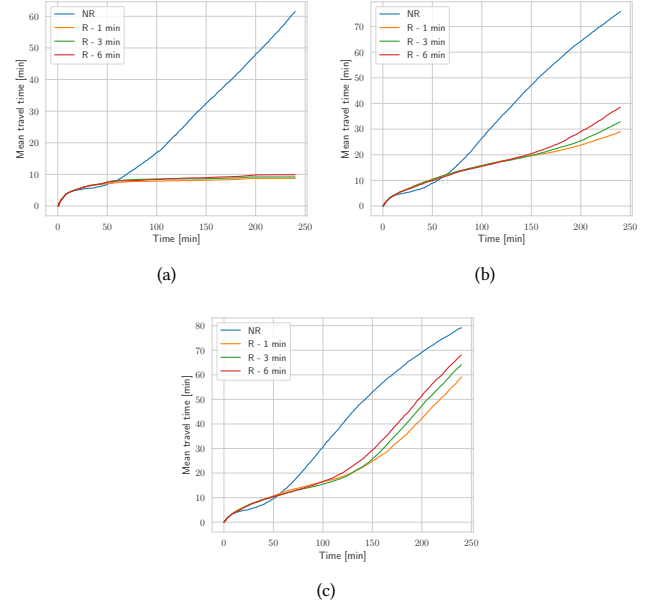


Figure 3: Re-routing impact on vehicles' mean trip time under different traffic demands: (a) Low traffic (40%), (b) Medium traffic (70%), (c) High traffic (100%).

50 minutes less to complete their trips compared to the NR case. In the NR case, vehicles select the shortest path to destination without considering the current traffic state (number of vehicles in streets). Therefore, at a specific time (minute 50 in Fig. 3(a)), when new vehicles enter the simulation, they will continue using already congested streets. This implies that the mean trip time and the number of vehicles running in the simulation increases. Before this time point, re-route actions are only triggered by spurious congestion situations, leading to slightly higher trip times than NR. The lower trip time by using rerouting when streets are continuously congested is because vehicles have updated information concerning the overall traffic conditions about the road map to make better decisions. Moreover, in the low traffic scenario, vehicles find alternative routes keeping mean trip time constant (steady situation).

We can see that as the re-routing interval increases, vehicles make worse decisions, especially under high traffic demands (rush hours), see Fig. 3(c). Further, by increasing the traffic demand in the scenario, the level of congestion increases consequently. We can see this behavior in Fig. 3(c) with 100% of the traffic demand. On the one hand, the NR case produces the highest trip times because it is more probable for a vehicle to get stuck in a congested situation. On the other hand, the improvement in terms of trip time due to vehicles' re-routing is less significant since roads in the road network begin to be blocked.

Trip length: In Fig. 4, the NR case (no vehicle can re-route its trip) produces the shortest trip lengths under the different traffic demands, even though vehicles take more time to complete their travels, as we see in Fig. 3. This is because, by default, SUMO looks for the fastest path from origin to destination, but without considering the presence of other vehicles in the map. Alternatively, in the R cases analysed, vehicles consider current traffic conditions to re-compute new optimal routes with respect to the travel time.

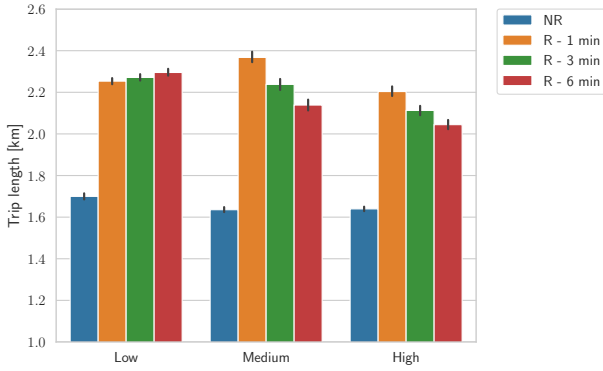


Figure 4: Impact of the re-routing capability on the vehicles' trip length for different traffic demands: Low (40%), medium (70%), and high (100%) of the maximum real traffic demand.

Each time a vehicle re-routes, it is because the new route takes less time than the current route (which suffers some level of congestion). Notice that the new route will be longer but with fewer cars on it. On the one hand, a vehicle that can re-route every 6 min will select longer paths in low-traffic conditions than the initial route planned. On the other hand, for Medium and High traffic, as the re-route interval increases, vehicles find a lower number of better routes than for the case $R=1$ min (see Fig. 5) and therefore, the average trip length decreases.

Route changes: Fig. 5 shows the mean number of route changes that vehicles perform per kilometer. As it is expected, the "R - 1 min" case produced more route changes. We can see that the effect of varying the re-routing interval is more significant under medium and high traffic conditions. Mainly, under a high traffic demand, using a re-route interval of 1 minute (R - 1 min), it doubles the number of route changes when compared to the "R - 3 min" case, and four times compared to the "R - 6 min" case.

We can conclude that re-routing capability modifies the mobility pattern making vehicle distribution more spread as vehicles choose different routes not congested (i.e., with the minimum number of vehicles). Therefore, re-routing reduces the overall vehicles' trip time. However, re-routing may introduce unrealistic mobility behavior during the simulation. For instance, Fig. 4 and Fig. 5 show that for the "R - 1 min" case, in a route of 2.2 km, vehicles would change their routes in average about 32 times (i.e., 16 re-routes/km). This is because roads along the map are congested (rush hour), so vehicles modify their routes very often. In real life, drivers would not change their routes so often (e.g., each minute), just when they are involved in a severe traffic jam.

4.3 Impact of vehicle re-routing capacity on vehicle network connectivity

To provide an idea of the impact of vehicle re-routing capacity in the connectivity of vehicular networks, we take advantage of graph theory concepts. First, we visually evaluate nodes' connectivity through the adjacency matrix obtained by enabling vehicles' re-routing. Then, through the use of well-know graph metrics [8], we intend to provide valuable insight into vehicular network behavior.

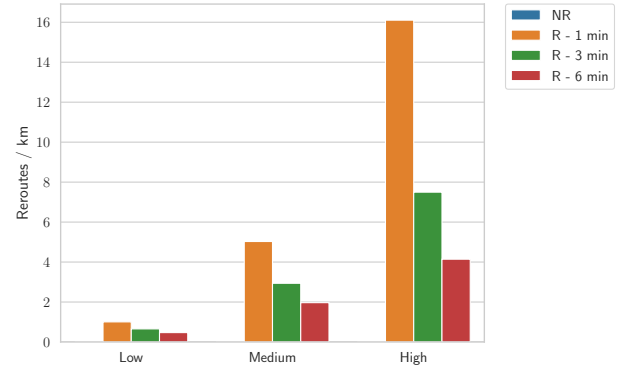


Figure 5: Number of route changes per kilometer for different traffic demands: Low (40%), medium (70%), and high (100%) of the maximum real traffic demand.

To this end, we consider the vehicular network settings summarized in Table 2.

Table 2: Vehicular network settings.

Parameters	Value
Transmission power	23 dbm
Sensitivity threshold	-82 dbm
Transmission range	400 m in line of sight (LOS)
Path loss model	Empirical IEEE 802.11p [9]

First, we assume all vehicles in the scenario are equipped with an IEEE 802.11p network card and have defined a transmission power of 23 dBm. Thus, the communication range will be approximately 400m. We also consider the building attenuation model proposed in [9] to model the non-line of sight (NLOS) condition to generate the building information, as it is described in Section 3. The first step to analyze the network is to convert vehicular traces in network snapshots at different simulation times. Then, the nodes' connectivity can be represented as a graph G where $V(G)$ refers to the set of *vertices* (nodes) of the graph, and $E(G)$ is the set of *edges* of the graph. An edge is added between two nodes (i, j) in case those nodes reach each other (i.e., reception power > -82 dBm). The network is represented by the adjacency matrix $(A_{i,j})$, where each entry corresponds to the weight of the edge $(E_{w_{i,j}})$. This weight of the edge corresponds to the signal strength in the link between two nodes, as follows:

$$A_{i,j} = \begin{cases} E_{w_{i,j}} & \text{if nodes } i, j \text{ are connected} \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

In Fig. 6, we depict the nodes' connectivity by means of the adjacency matrix. To obtain the edges' weight of the graph, we have added the signal strength each time that two nodes are in contact during the period of study. We take into account the 250 nodes with the highest signal strength. The connection intensity varies from light-red (few nodes are connected) to very dark-red (many nodes are connected), depending on the node's edge weights. We evaluate two cases considering low (off-peak hours) and high (peak hours) traffic demands. In the figure, R cases mean that all

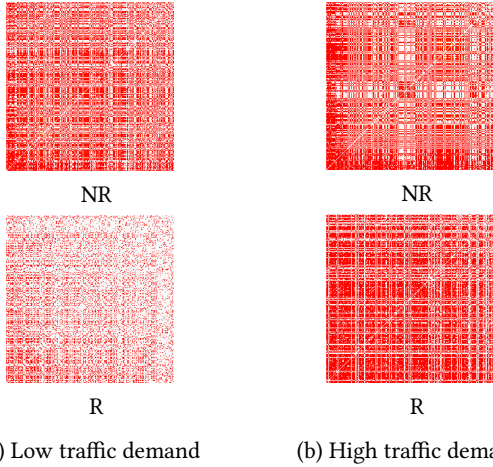


Figure 6: Adjacency matrices for vehicular traces. Light-red (dark-red) color depicts low (strong) nodes' connectivity. R (NR) means all (none) vehicles can re-route their trips

vehicles in the simulation can perform route changes during their trips, whereas *NR* cases mean that no re-routing is allowed. We use below the terms *vehicle* and *node* interchangeably.

Fig. 6(a) shows that under a low traffic demand, the *NR* case obtains a set of nodes that are highly interconnected (there are few white spaces), while the *R* case represents a sparser nodes' connectivity. The reason is that in case *R* vehicles are allowed to choose alternative routes to avoid congested roads. Also, vehicles move faster, reducing connectivity opportunities. This way, we can notice how the re-routing capacity of vehicles affects negatively the vehicular network connectivity under low traffic demands.

In contrast, under high traffic demands, see Fig. 6(b), the *R* case shows a large amount of connected nodes with high intensity. The reason is that as the traffic demand increases, the probability of a vehicle to be in a traffic jam increases too. In such a case, vehicles would move slowly (with higher trip times, see Fig. 3(c)) so that nodes increase their connectivity opportunities. However, for very high traffic demands traffic jams are unavoidable, which could also lead to a congested vehicular network with a high packet collision probability.

In the rest of this section, we analyze the nodes' connectivity in terms of well-known graph metrics to characterize vehicular networks. For this, we define two scenarios called *Average*, and *Summary*, as follows:

Average graph: We construct a graph G_t for every snapshot. Then, we average the metrics computed for each graph separately. We generate a network graph every 30s, where the edges' weight is the strength of the signal. Recall that an *edge* of the graph links two *vertices* (vehicles) are in the transmission range of each other. This metric provides an idea of the behavior of the vehicular network at a specific moment.

Summary graph: We construct a single *summary* graph G considering all the samples during a period. In case two nodes were in the transmission range of each other (at least once during the simulation), we draw an edge between them. Weights of edges are obtained by adding the signal strength each time two nodes are in

contact. This metric gives an idea of how often two nodes coincide during the studied period.

Edge percentage (EP): The EP is defined in the range $[0, 1]$, where $EP=0$ means no connectivity between nodes and $EP=1$ means full connectivity between nodes. The EP is defined as $EP = \frac{E}{N \cdot (N-1)}$. Here, E is the number of edges and N the total number of vertices (nodes) of G . Fig. 7 shows the *average* and *summary* graphs detailed above. Regarding the *average* graph, all re-routing cases produce a similar behavior. The ED is low, just 1,3% of vehicles are (directly) connected under sparse scenario (Low traffic); furthermore, this index halves under high traffic condition. This effect is more notable when re-routing is allowed, since vehicles can modify their planned routes to avoid congested streets so they will find less vehicles in the new routes. Regarding the *summary* graph, in case of low traffic

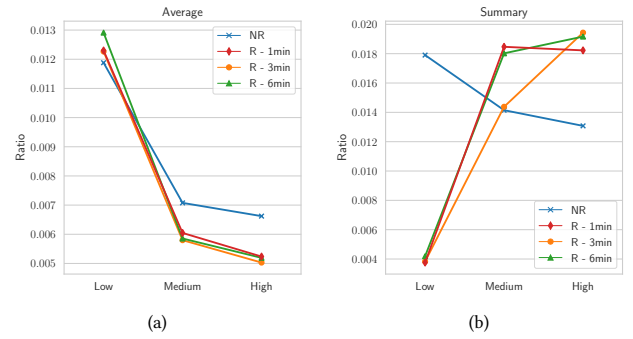


Figure 7: Edge percentage (EP) to measure the connectivity level between nodes (vehicles). (a) Average graph, (b) summary graph.

demand by enabling re-routing capabilities (*R* cases) nodes' connectivity between nodes is highly affected (negatively) compared to the *NR* case (no re-routing is allowed). This is because during the studied period, vehicles select new routes with low congestion levels so that vehicles are sparsely distributed and consequently the EP decreases. In contrast, with high traffic demand, the EP values increase (to a higher value than the *NR* case), meaning that it is possible to establish more connections between nodes compared to the *NR* case (no re-routing is allowed). The reason is that vehicles looking for a new route will try to look for a faster route, but also with a lot of traffic. The results of the *summary* graph correspond with observed behavior with the adjacency matrices in Fig. 6. That is, the re-routing capacity of vehicles negatively affects the vehicular network connectivity under low traffic demands.

Transitivity (T): The transitivity of a graph is defined as the ratio of *triangles* in the graph (i.e., $a \rightarrow b, b \rightarrow c, c \rightarrow a$), compared to the total number of *connected triples* (i.e., $a \rightarrow b, b \rightarrow c$). T indicates the proportion of nodes that have adjacent nodes interconnected. It is defined in the range $[0, 1]$, where $T=1$ if the network contains all possible edges (full connectivity). In opportunistic networks, transitivity can be interpreted as the chances that two nodes that communicate directly ($a \rightarrow b$) can also communicate back by using only two additional hops ($b \rightarrow c, c \rightarrow a$). In Fig. 8, the higher associated transitivity takes place with *NR* under low and medium traffic. We can conclude that by enabling re-routing capabilities, less back-up triangle paths will be generated. This is due to the fact that

dynamic route planning diminishes congestion levels throughout the road map. In this case, vehicles move faster (see, Fig. 3), so that it is less probable that back-up triangles can be generated.

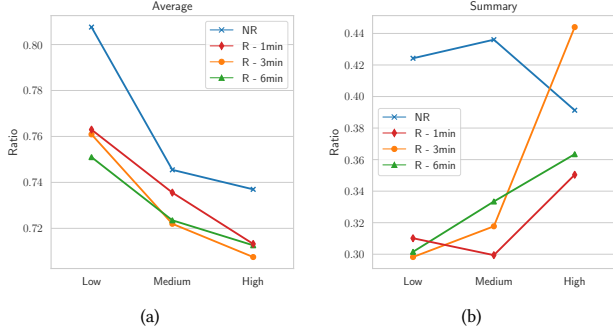


Figure 8: Normalized transitivity (T), where $T=1$ if the network contains all possible edges (full nodes' connectivity).

Diameter (D): The diameter of a graph G is the length of the longest shortest path (obtained in terms of the number of hops) between any two graph vertices (nodes) i and j in G . This way, D reflects the largest number of vertices which must be traversed in order to travel from one vertex to another (paths which back-track, detour or loop are not considered). Regarding the *average* scenario in Fig. 9, we can see that as the traffic demand increases, diameter values for R-cases (re-routing is allowed) also increase. In the context of vehicular communications, a long diameter value means that distant nodes could be reached using a suitable routing protocol. This result agrees to the one obtained for the edge percentage, see Fig. 7. A low EP value means that more hops are needed to reach distant nodes. Regarding the *summary* case, we can see that for low traffic density, as the re-routing interval decreases (i.e., re-routing events occurs more often), the diameter increases. Particularly, in the "R - 1 min" case, vehicles cover more areas of the map (vehicles are spread distributed), so that vehicles will require more hops to connect to distant vehicles. In case of medium and high traffic demands, no significant difference is produced for the three re-routing intervals considered.

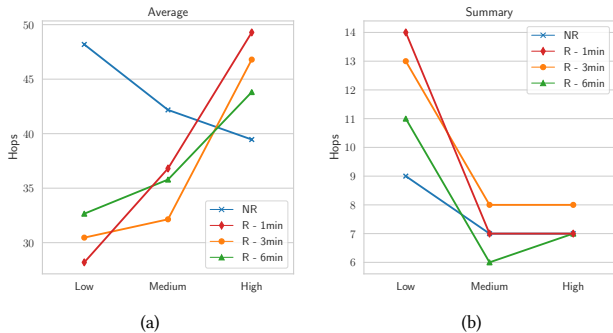


Figure 9: Diameter (D). Large D values mean vehicles will require more hops to connect to distant vehicles.

Mean node's degree: The degree of a node i is defined as $deg(i) = |N(i)|$, where $N(i)$ is the number of neighbors of that node. In Fig. 10, as it is expected when the traffic demand increases the

average nodes' degree increases as well (i.e., nodes have a higher number of neighbors). Focusing on the *average* case, we can see that while the NR case obtains higher node degree values when traffic demand is low and medium, with high traffic demand all cases behave similarly. Regarding the *summary* case, re-routing options produce higher values for medium and high traffic demands. It means that nodes will have more options to forward packets under high traffic demands for all cases. However, very high traffic demands can lead to congested roads (traffic jams) and also to congested vehicular network.

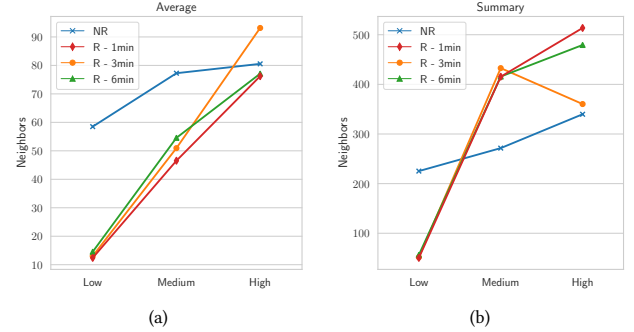


Figure 10: Mean nodes' degree. It shows the average number of neighbors.

Betweenness centrality: It measures the number of shortest paths that go through a particular node. This metric shows if there are nodes that often become intermediate nodes between other nodes in the network. We use the signal strength (i.e., edges' weight) to compute the shortest paths. In Fig. 11, a high value shows that a set of nodes arises repetitively in the shortest path creation. We can see that the NR option (no re-routing is allowed) reaches higher values for medium and high traffic demands. This is because vehicles mainly follow the same routes in their trips (computed in an empty road map) so that it produces similar intermediate nodes. On the other hand, among the "R cases" (re-routing is allowed), there is not a significant difference, since vehicles choose different routes based on traffic conditions. It is important to notice that the centrality score accounts for the disproportion of the metric (in this case betweenness) among nodes of the graph. In this regard, medium traffic load leads to the highest disproportion in the score of nodes. However, the average betweenness value of a node in high traffic is higher than in the medium traffic case.

Number of Communities: This metric shows the tendency of a graph with nodes densely connected among each other, but sparsely connected to other communities. This metric is useful to infer how clusters are formed when routing protocols based on cluster-approach work. To detect the communities, we use the algorithm proposed in [3] for every graph. In Fig. 12, we can see that a higher number of clusters are obtained for "R cases", especially when vehicles are more sparsely distributed. As it is expected, the number of clusters decreases as the traffic demand increases since more often vehicles encounter each other during the simulation.

5 CONCLUSIONS

In this work, we have studied the impact of enabling dynamic route planning in vehicular simulations, from two points of view: (i) how

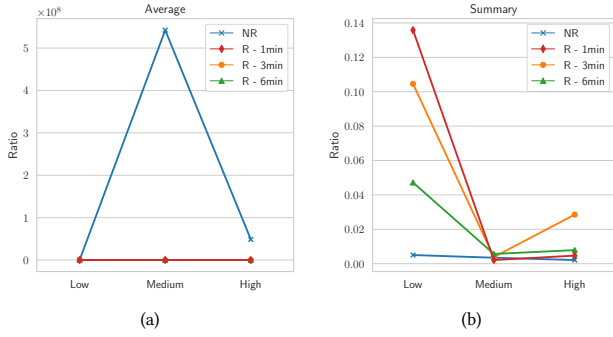


Figure 11: Betweenness centrality. It highlights if there are nodes that often become intermediate nodes between other nodes in the network.

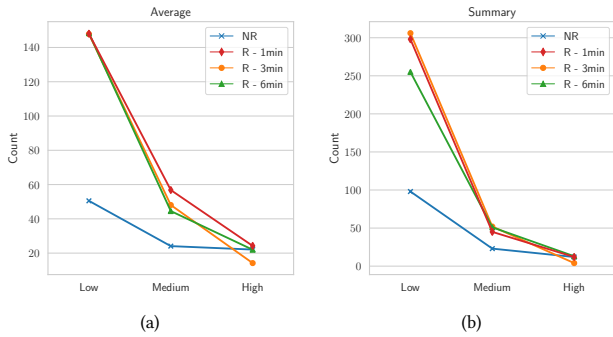


Figure 12: Number of communities. It shows the tendency to form clusters in the network.

re-routing in vehicles, which is interesting to have more realistic simulations, affect the vehicle's mobility; and (ii) how re-routing affect the nodes' connectivity in the vehicular network.

On the one hand, the first analysis shows that re-routing capabilities modifies the shape of the vehicular traffic in the roads, minimizing traffic congestion levels and making vehicles' distribution more spread in the roads. In real life, drivers use re-routing to look for faster paths till destination and also to be able to avoid congested roads. Effectively, by allowing re-routing in our simulations we have measured lower vehicles' trip time. Moreover, enabling re-routing capabilities helps to avoid unrealistic situations like congestion under low-traffic load due to the SUMO's default configuration of static routes. Therefore, considering re-routing is essential to produce meaningful evaluation results when assessing ITS services. On the other hand, we have analyzed the nodes' connectivity by using well-known graph metrics for network analysis. Results show that by enabling re-routing capabilities, vehicular network connectivity is highly impacted specially in case of low traffic demands (off-peak hours). At the same time, it improves the connectivity of the vehicular network under high traffic demands (peak hours). Concluding, SUMO vehicular traces with re-routing capabilities would significantly modify the simulation results obtained when we assess the performance of services, such as traffic reporting and accident warnings. This fact has paramount importance to obtain realistic simulation results.

Allowing re-routing in vehicular scenarios is of paramount importance to attain realistic simulations. For instance, varying too often, the re-routing frequency may generate too many route changes, which may introduce anomalies during the simulation (i.e., unrealistic behavior). In a future work, we will test a more granular set of R values and of traffic update intervals. Also, we will use different traffic demand generators. Additionally, we will assess a more extensive map area, including accidents during the simulation, to analyze the impact of vehicles re-routing their trips to avoid the area around the accident.

ACKNOWLEDGMENTS

This work was supported by the Spanish Government under research project "sMART Grid using Open Source intelligence" (MAGOS) TEC2017-84197-C4-3-R. Pablo Barbecho is the recipient of a grant from Secretaría Nacional de Educación Superior, Ciencia y Tecnología SENESCYT.

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